

Unit 19a: Glaciers, Glaciation & Glacial Erosion

Reading: pages 274–287 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. A glacier is defined as a long-lasting body of ice that is large enough to
 - a) move under its own weight
 - b) cover an entire continent
 - c) reach all the way to the ocean
 - d) survive through a single summer
2. Glaciers hold approximately what share of all the freshwater on Earth?
 - a) About 10%
 - b) About 30%
 - c) About 69%
 - d) About 95%
3. The intense late Proterozoic glaciations of the Cryogenian Period are nicknamed
 - a) the Karoo glaciations
 - b) the Wisconsin glaciations
 - c) the Huronian glaciations
 - d) the “Snowball Earth” glaciations
4. Which tectonic event at about 40 Ma allowed the Antarctic Circumpolar Current to form and cool Antarctica?
 - a) The opening of the Drake Passage between South America and Antarctica
 - b) The closing of the connection between North and South America
 - c) The breakup of the supercontinent Gondwana
 - d) The collision of India with Asia
5. The boundary between a glacier’s zone of accumulation and its zone of ablation is called
 - a) the terminal moraine
 - b) the equilibrium line
 - c) the crevasse field
 - d) the nunatak

6. Glacial ice behaves plastically and begins to flow below a depth of roughly
- a) 5 to 10 m
 - b) 50 to 100 m
 - c) 500 m
 - d) 4,000 m

Part B · True or False

2 questions × 1 point = 2 points

7. Cool summers promote glacial advance, and summer melting matters more to a glacier's budget than winter snowfall.

TRUE / FALSE

8. Glacial ice is extremely hard, and the ice itself does most of the eroding beneath a glacier.

TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Explain how falling snow is transformed into glacial ice.

10. Name and describe four landforms produced by alpine glacial erosion.

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